WORKED SOLUTION

STEP II 1994, Question 7

Pure • Coordinate geometry

STEP II 1994, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q7

The key idea

The angle of elevation of the top of a pole of height h seen from distance r has tangent h/r, so "equal angles of elevation" just says that the soldier's distances from the two feet are in the fixed ratio $h_1 : h_2$. Squaring that condition gives an equation of exactly the form in the first part, so the locus is a circle (a circle of Apollonius). For the last part, put the poles at general positions and write each centre as a weighted combination of two feet; the identity then drops out, and its coefficients add up to zero, which is what forces collinearity.

Solution

The circle

Since $a > 0$ we may divide by a and complete the square:

$$x^2 + y^2 + \frac{2g}{a}x + \frac{2f}{a}y + \frac{c}{a} = 0 \iff \left(x + \frac{g}{a}\right)^2 + \left(y + \frac{f}{a}\right)^2 = \frac{g^2 + f^2 - ac}{a^2}.$$

The right-hand side is positive because $f^2 + g^2 > ac$, so this is a circle with

$$\text{centre } \left(-\frac{g}{a}, -\frac{f}{a}\right), \qquad \text{radius } \frac{\sqrt{g^2 + f^2 - ac}}{a}.$$

(Without the condition the "circle" would be a single point or empty.)

Two flagpoles

Put the foot of the pole of height h_1 at the origin and the foot of the other at $(d, 0)$. If the soldier is at (x, y), at distances r_1 and r_2 from the two feet, the angles of elevation θ_1, θ_2 satisfy $\tan\theta_i = h_i/r_i$. Both angles lie in $(0, \frac{\pi}{2})$, where tan is one-to-one, so

$$\theta_1 = \theta_2 \iff \frac{h_1}{r_1} = \frac{h_2}{r_2} \iff h_2^2 r_1^2 = h_1^2 r_2^2,$$

the last step being reversible because all the quantities are positive. (The soldier can't stand at a foot: $r_1 = 0$ would force $r_2 = 0$ too.) So the path is

$$h_2^2(x^2+y^2) = h_1^2((x-d)^2+y^2) \iff (h_2^2-h_1^2)(x^2+y^2) + 2h_1^2 d\,x - h_1^2 d^2 = 0.$$

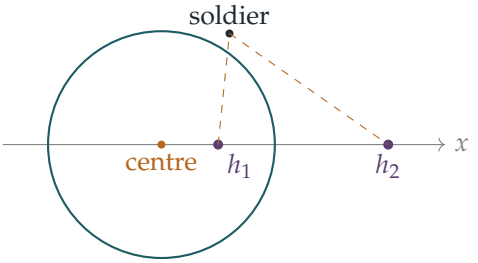
Label the poles so that $h_2 > h_1$ (if they're different, one is taller). Then this is the first part's equation with

$$a = h_2^2 - h_1^2 > 0, \quad g = h_1^2 d, \quad f = 0, \quad c = -h_1^2 d^2,$$

and

$$g^2 + f^2 - ac = h_1^4 d^2 + (h_2^2 - h_1^2)h_1^2 d^2 = h_1^2 h_2^2 d^2 > 0.$$

So the soldier marches round a circle with centre $\left(-\dfrac{h_1^2 d}{h_2^2 - h_1^2},\, 0\right)$ and radius $\dfrac{h_1 h_2 d}{h_2^2 - h_1^2}$. The centre lies on the line through the feet, beyond the shorter pole. The circle goes round the shorter pole, which makes sense: to make the short pole look as tall as the tall one, you have to stand nearer to it.



(Drawn for $h_1 = 1$, $h_2 = 2$, $d = 3$: the centre is $(-1, 0)$ and the radius is 2.)

If $h_1 = h_2$, the $x^2 + y^2$ terms cancel and the equation becomes $2h_1^2 d\,x = h_1^2 d^2$, that is $x = \frac{1}{2}d$. The soldier walks along the perpendicular bisector of the line joining the feet, a straight line rather than a circle. This is the limit of the circles above: as $h_2 \to h_1$ the radius tends to infinity.

Three flagpoles

Now we can't put all three feet on an axis, so let the pole of height h_i stand at $\mathbf{p}_i = (p_i, q_i)$. Repeating the argument above with $r_i^2 = (x-p_i)^2 + (y-q_i)^2$, the soldier assigned to poles i and j walks along

$$(h_j^2 - h_i^2)(x^2+y^2) - 2(h_j^2 p_i - h_i^2 p_j)x - 2(h_j^2 q_i - h_i^2 q_j)y + \text{const} = 0.$$

This is a circle (the value of $g^2 + f^2 - ac$ doesn't change when we move the origin, so it is still $h_i^2 h_j^2 |\mathbf{p}_i - \mathbf{p}_j|^2 > 0$, after multiplying through by -1 if $h_j < h_i$). Using the first part, its centre $\mathbf{c}_{ij} = (x_{ij}, y_{ij})$ satisfies

$$(h_j^2 - h_i^2)\,\mathbf{c}_{ij} = h_j^2 \mathbf{p}_i - h_i^2 \mathbf{p}_j.$$

This expression doesn't change if we swap i and j (both sides change sign), so it doesn't matter which way round we label $\mathbf{c}_{31}$.

Ask yourself How do we turn three of these formulae into the given identity without solving for anything?

Each term in the identity already has a factor $\pm(h_j^2 - h_i^2)$ next to x_{ij}, so we can substitute the formula above directly, using vectors so that the x- and y-coordinates are done together:

$$\begin{aligned} h_3^2(h_1^2 - h_2^2)\,\mathbf{c}_{12} &= -h_3^2(h_2^2\mathbf{p}_1 - h_1^2\mathbf{p}_2) = -h_2^2h_3^2\,\mathbf{p}_1 + h_1^2h_3^2\,\mathbf{p}_2, \\ h_1^2(h_2^2 - h_3^2)\,\mathbf{c}_{23} &= -h_1^2(h_3^2\mathbf{p}_2 - h_2^2\mathbf{p}_3) = -h_1^2h_3^2\,\mathbf{p}_2 + h_1^2h_2^2\,\mathbf{p}_3, \\ h_2^2(h_3^2 - h_1^2)\,\mathbf{c}_{31} &= h_2^2(h_3^2\mathbf{p}_1 - h_1^2\mathbf{p}_3) = h_2^2h_3^2\,\mathbf{p}_1 - h_1^2h_2^2\,\mathbf{p}_3. \end{aligned}$$

Adding, everything cancels in pairs, so

$$h_3^2(h_1^2 - h_2^2)\,\mathbf{c}_{12} + h_1^2(h_2^2 - h_3^2)\,\mathbf{c}_{23} + h_2^2(h_3^2 - h_1^2)\,\mathbf{c}_{31} = \mathbf{0}.$$

The x-component is the required relation between the x_{ij}, and the y-component is the same relation between the y_{ij}.

Ask yourself Why does a linear relation between three points make them collinear?

Call the three coefficients $A = h_3^2(h_1^2 - h_2^2)$, $B = h_1^2(h_2^2 - h_3^2)$, $C = h_2^2(h_3^2 - h_1^2)$. The heights are positive and all different, so none of A, B, C is zero. The key extra fact is that they add up to zero:

$$A + B + C = h_1^2h_3^2 - h_2^2h_3^2 + h_1^2h_2^2 - h_1^2h_3^2 + h_2^2h_3^2 - h_1^2h_2^2 = 0.$$

So $C = -A - B$, and the relation becomes

$$A(\mathbf{c}_{12} - \mathbf{c}_{31}) + B(\mathbf{c}_{23} - \mathbf{c}_{31}) = \mathbf{0}, \qquad \text{so} \qquad \mathbf{c}_{12} - \mathbf{c}_{31} = -\frac{B}{A}\,(\mathbf{c}_{23} - \mathbf{c}_{31}).$$

The vectors from $\mathbf{c}_{31}$ to the other two centres are parallel (or one is zero, if two centres coincide), so the three centres lie on a straight line. Without $A + B + C = 0$ a linear relation would prove nothing: any three points satisfy some relation of that kind.

Checks

- With $h_1 = 1$, $h_2 = 2$, $d = 3$ our formulae give centre $(-1, 0)$ and radius 2. The point $(1, 0)$ is at distances 1 and 2 from the feet, and $(-3, 0)$ is at distances 3 and 6: both in ratio $1 : 2$, as they should be.
- A numerical check put random points on each circle and confirmed that the two angles of elevation agreed. For 2000 random triples of poles in random positions, it also confirmed the identity and the collinearity of the three centres.
- Another reason to expect collinearity: if a point lies on two of the circles, then $r_1 : r_2 = h_1 : h_2$ and $r_2 : r_3 = h_2 : h_3$, so $r_1 : r_3 = h_1 : h_3$ and it lies on the third. When the circles meet, they all pass through the same two points, so their centres all lie on the perpendicular bisector of those points.

Common slips

- Setting $h_1/r_1 = h_2/r_2$ without saying why equal angles give equal tangents (both angles are acute).
- Forgetting to check $g^2 + f^2 > ac$: the first part only gives a circle under that condition.
- Deducing collinearity from the identity alone, without using $A + B + C = 0$ and $A, B, C \neq 0$.

Beyond STEP

The set of points whose distances from two fixed points are in a fixed ratio $k \neq 1$ is a *circle of Apollonius*. Taking all values of k gives a *coaxal system* of circles, a family in which any two circles share the same radical axis. Families like this are where inversive geometry begins, and in the complex plane they are the circles $|z - a| = k|z - b|$, which Möbius transformations map to one another. Coxeter and Greitzer, *Geometry Revisited*, covers coaxal circles and inversion; Needham, *Visual Complex Analysis*, shows the complex-number side.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.